

Practical *De Novo* Nanobody Discovery with Tens of Experimental Candidates

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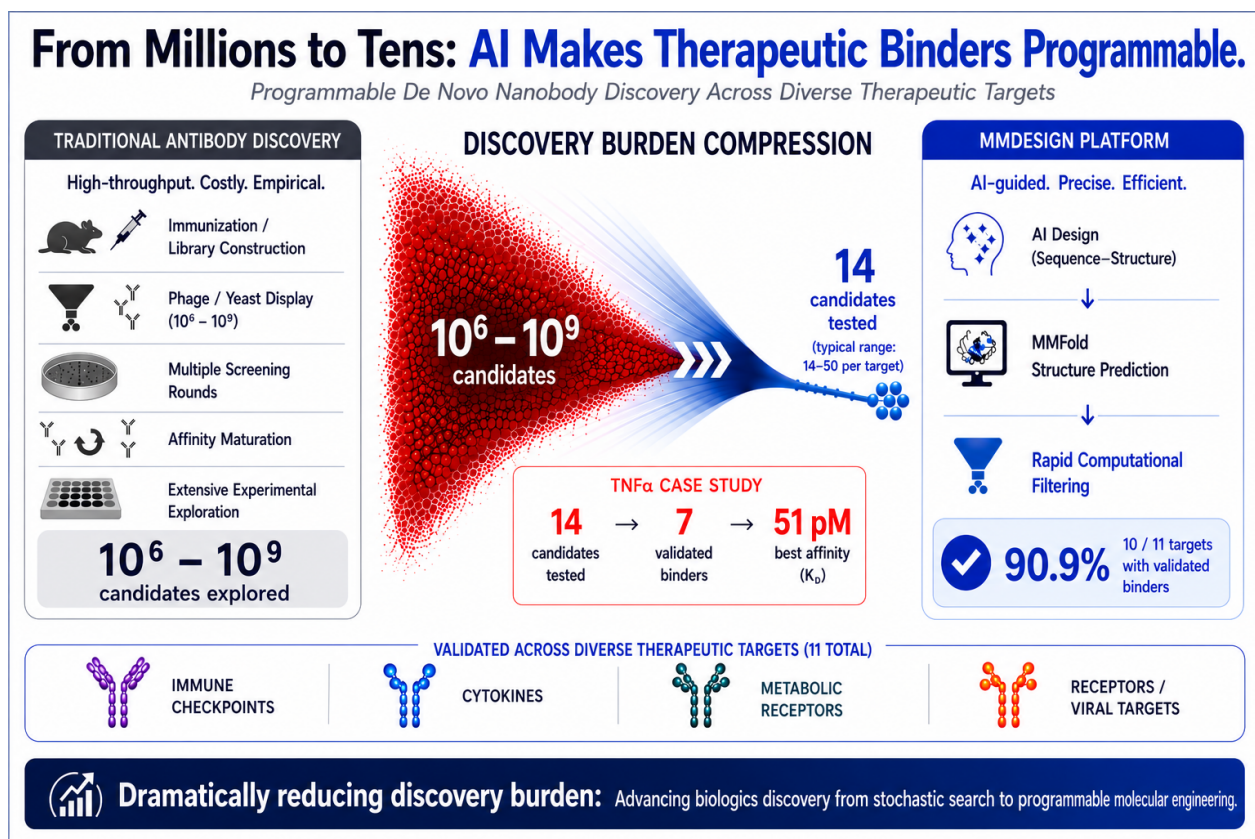
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Abstract

Conventional antibody discovery relies on animal immunization or large-scale library screening, requiring experimental interrogation of millions to billions of candidates with limited programmability over epitope specificity and molecular properties. Recent advances in generative modeling have demonstrated the possibility of *de novo* antibody design, yet achieving experimentally practical biologics discovery at low experimental throughput across therapeutically relevant targets remains challenging.

Here we present **MMDesign**, a *de novo* VHH discovery platform combining generative sequence–structure optimization with multi-tier computational filtering. Starting from only a target protein and specified epitope residues, MMDesign generates tens of thousands of nanobody candidates through optimization over the confidence landscape of **MMFold**, a proprietary antibody–antigen structure prediction model, together with a protein language model. Successive filtering stages incorporating structural reliability, sequence naturalness, and physics-based interface evaluation compress the candidate pool to only tens of experimental testable designs per target. We evaluated MMDesign across 11 therapeutically relevant targets spanning cytokines, immune checkpoints, receptors, a viral protein, and a surface antigen. Experimental testing of only 14–50 candidates per target yielded confirmed binders for **10 of 11 targets (90.9%)**, with best measured affinities ranging from picomolar to nanomolar levels, demonstrating broad target-level success at experimentally tractable scale. Notably, MMDesign achieved a 50% hit rate on TNF α , a shallow trimeric cytokine target that has proven difficult for prior low-throughput *de novo* VHH discovery campaigns.

Together, these results suggest that biologics discovery may increasingly transition from large-scale stochastic screening toward programmable molecular engineering, in which computational generation and prioritization substantially reduce experimental burden while enabling rapid epitope-targeted therapeutic binder discovery.



Graphical Abstract.

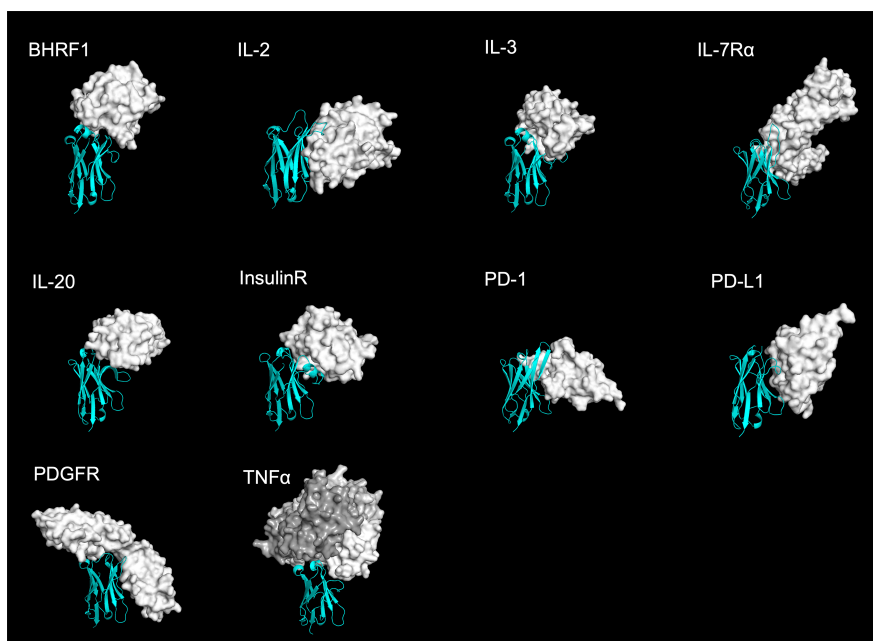


Figure 1: Predicted structures of the top-ranked VHH binder (cyan) in complex with each of the ten successful targets (white).

1 Introduction

Therapeutic antibody discovery remains dominated by large-scale stochastic screening, including animal immunization and display-library selection, often requiring experimental interrogation of millions to billions of variants. Although these approaches have produced numerous successful therapeutics, they offer limited control over epitope specificity and therapeutic properties as well as impose substantial experimental and temporal costs. A central challenge in computational biologics engineering is therefore whether antibody discovery can transition from stochastic screening toward programmable molecular generation and optimization.

Recent advances in generative modeling have transformed *de novo* protein binder design, with systems such as RFdiffusion [1], BindCraft [2], and AlphaProteo [3] demonstrating that high-affinity binders can be computationally generated directly from target structure. Extending these advances to antibodies, however, has proven substantially more difficult because antibody-antigen recognition is dominated by conformationally flexible CDR loops and highly heterogeneous binding interfaces. Nevertheless, recent systems including Chai-2 [4], Germinal [5], BoltzGen [6], and GeoFlow-V3 [7] have begun to demonstrate proof-of-concept *de novo* antibody and nanobody discovery.

Despite these advances, practical low-throughput antibody discovery remains unresolved. Most current systems exhibit inconsistent success across targets, require substantial experimental exploration, or have yet to demonstrate robust performance on difficult therapeutic antigens. In particular, few studies have reported broad target-level success while experimentally testing only tens of candidates per target.

Here we present **MMDesign**, a hallucination-guided *de novo* VHH discovery platform designed to operate at experimentally tractable scales. Starting from only a target protein and specified epitope residues, MMDesign combines sequence and structure co-generation with multi-tier computational filtering to produce only tens of experimental candidates per target. We systematically evaluated MMDesign across 11 therapeutically relevant targets spanning cytokines, immune checkpoints, receptors, a viral protein, and a surface antigen. Experimental testing of only 14–50 candidates per target yielded confirmed binders for 10 of 11 targets (90.9%), with best measured affinities ranging from picomolar to nanomolar levels. Notably, MMDesign achieved a 50% hit rate on TNF α , a trimeric cytokine target that has proven difficult for prior low-throughput *de novo* VHH discovery efforts [4, 6]. Together, these results suggest that generative nanobody discovery is beginning to transition from proof-of-concept demonstrations toward a practical therapeutic discovery modality.

2 Results

2.1 Practical De Novo VHH Discovery Across Therapeutically Diverse Targets

To evaluate whether generative nanobody discovery can operate at experimentally practical scales, we systematically evaluated MMDesign across 11 therapeutically relevant targets while experimentally testing only 14–50 candidates per campaign.

Target diversity and structural complexity of the benchmark panel. The target panel evaluated in this study spans a broad range of therapeutically relevant protein classes and represents one of the most structurally and biologically diverse benchmark sets reported for recent *de novo* antibody or nanobody design systems. The eleven targets include cytokines (TNF α , IL-2, IL-3, IL-20), immune checkpoints (PD-1, PD-L1), receptor ectodomains (InsulinR, PDGFR, IL-7R α), a viral protein (BHRF1), and a surface antigen (CD7), collectively covering immunology, oncology, metabolism,

and infectious disease applications. Unlike many recent generative antibody-design studies that focus primarily on a limited set of demonstration antigens, the MMDesign panel more closely resembles the diversity encountered in real therapeutic biologics discovery.

Importantly, the benchmark diversity extends beyond biological categories and includes substantial variation in interface geometry and binding difficulty. $\text{TNF}\alpha$ represents a particularly challenging case because it forms a homotrimeric cytokine complex with a shallow and solvent-exposed binding surface, a target class that has historically proven difficult for low-throughput generative binder design. Receptor targets such as InsulinR, PDGFR, and IL-7R α introduce additional complexity through their large, flexible ectodomains and heterogeneous surface topologies. Together, these targets span multiple binding regimes, including shallow exposed grooves, flexible receptor interfaces, checkpoint interaction surfaces, and viral epitopes. As a result, the MMDesign benchmark evaluates not only whether the platform can generate binders, but also whether it can generalize across structurally distinct and therapeutically realistic antigen classes.

Candidate VHHs were generated through hallucination-guided sequence–structure optimization followed by multi-tier computational filtering incorporating structural reliability analysis, sequence naturalness evaluation, and physics-based interface scoring.

Across the 11 campaigns, confirmed specific binders were obtained for **10 targets (90.9%)**. Experimental testing involved only tens of candidates per target, yet yielded per-target hit rates ranging from 2.3% to 86.7% (Table 1). Best measured affinities ranged from picomolar to micromolar levels, with several campaigns producing binders in the low- to mid-nanomolar regime. In all successful campaigns, predicted complex structures indicated direct contacts between designed CDR residues and the user-specified epitope hotspot residues.

Among all targets, PD-L1 achieved the highest hit rate, with 26 of 30 tested candidates confirmed as specific binders and a best measured K_D of 7.2 nM. Additional campaigns yielded multiple confirmed binders across diverse antigen classes, including InsulinR (5/25, best K_D 25 nM), BHRF1 (5/35, best K_D 282 nM), IL-2 (5/50, best K_D 138 nM), and IL-3 (2/25, best K_D ~200 nM). Single confirmed binders were also identified for PDGFR, IL-7R α , IL-20, and PD-1. Only one target, CD7, yielded no confirmed binders under the current pipeline configuration.

Target	Category	Hits / Tested	Hit Rate (%)	Best K_D
PD-L1	Checkpoint	26 / 30	86.7	7.2 nM
$\text{TNF}\alpha$	Cytokine	7 / 14	50.0	51 pM*
InsulinR	Receptor	5 / 25	20.0	25 nM
BHRF1	Viral protein	5 / 35	14.3	282 nM
IL-2	Cytokine	5 / 50	10.0	138 nM
IL-3	Cytokine	2 / 25	8.0	~200 nM
PDGFR	Receptor	1 / 25	4.0	150 nM
IL-7R α	Receptor	1 / 25	4.0	408 nM
IL-20	Cytokine	1 / 40	2.5	6.5 μ M
PD-1	Checkpoint	1 / 44	2.3	38 nM
CD7	Surface antigen	0 / 25	0	—

*Apparent K_D reflects avidity due to the trimeric nature of $\text{TNF}\alpha$ analyte.

Table 1: Experimental validation results across all 11 target campaigns. Hit Rate = confirmed hits / candidates tested. Best K_D measured by BLI unless otherwise noted.

Together, these results demonstrate that MMDesign can achieve broad target-level success while experimentally testing only tens of candidates per target, suggesting that generative VHH discovery may increasingly operate at experimentally tractable scales.

Structural analysis of the designed complexes indicates these binders are genuinely novel: 9 of the 11 targets have no known VHH complex in the PDB, and the designs occupy previously uncharacterized binding-pose space (Appendix C). At the same time, for targets where validated references exist, the designs converge on the same functional epitopes despite entirely *de novo* CDRs—including recovery of the natural PD-1:PD-L1 inhibitory interface.

2.2 MMDesign Achieves a 50% Hit Rate on the Difficult TNF α Target

TNF α is a homotrimeric cytokine whose shallow and solvent-exposed binding surface has historically proven difficult for low-throughput *de novo* VHH discovery approaches [4, 6]. Prior generative nanobody campaigns reported in the literature have failed to produce confirmed binders against this target under similarly limited experimental throughput settings.

To evaluate MMDesign on this challenging system, all three CDR loops were optimized through the hallucination-guided generation pipeline, and 14 candidates were selected for experimental characterization. Seven designs (50%) exhibited clear and specific target binding by BLI. The lead molecule achieved an apparent K_D of 51 pM, noting that this value reflects avidity enhancement associated with the trimeric TNF α analyte.

The TNF α campaign is particularly notable because it demonstrates substantial enrichment under extremely limited experimental exploration. Rather than requiring large-scale stochastic screening, the computational pipeline successfully narrowed tens of thousands of generated candidates to a small experimentally tractable set enriched for true binders. These results suggest that AI-guided generative discovery may increasingly access difficult therapeutic geometries that have historically resisted low-throughput discovery workflows.

Representative predicted structures and BLI sensorgrams for selected targets are shown in Figure 2.

2.3 Designed VHHs Exhibit Favorable Preliminary Developability Characteristics

In addition to binding activity, many designed VHHs exhibited encouraging preliminary developability-related properties. Expression yields across campaigns demonstrated that numerous candidates could be robustly expressed as VHH-Fc constructs in CHO cells (Figure 3). Size-exclusion chromatography further showed that many purified candidates remained predominantly monomeric with limited aggregation signatures (Figure 4).

To assess preliminary specificity characteristics, a subset of designed VHHs was evaluated for off-target binding against human serum albumin (HSA), the most abundant protein in plasma and a common source of non-specific interactions in therapeutic antibody development. Across all tested candidates, BLI responses against HSA remained minimal at micromolar concentrations, with no measurable evidence of strong non-specific binding (Figure 5). Additional poly-reactivity ELISA experiments against insulin and dsDNA similarly indicated low levels of broad non-specific reactivity.

Although these experiments do not constitute a comprehensive therapeutic developability assessment, they suggest that MMDesign can generate experimentally tractable antibody-like molecules rather than merely isolated binding sequences. These observations suggest that future generative biologics systems may increasingly optimize simultaneously for binding activity, specificity, expression, stability, and downstream therapeutic developability characteristics.

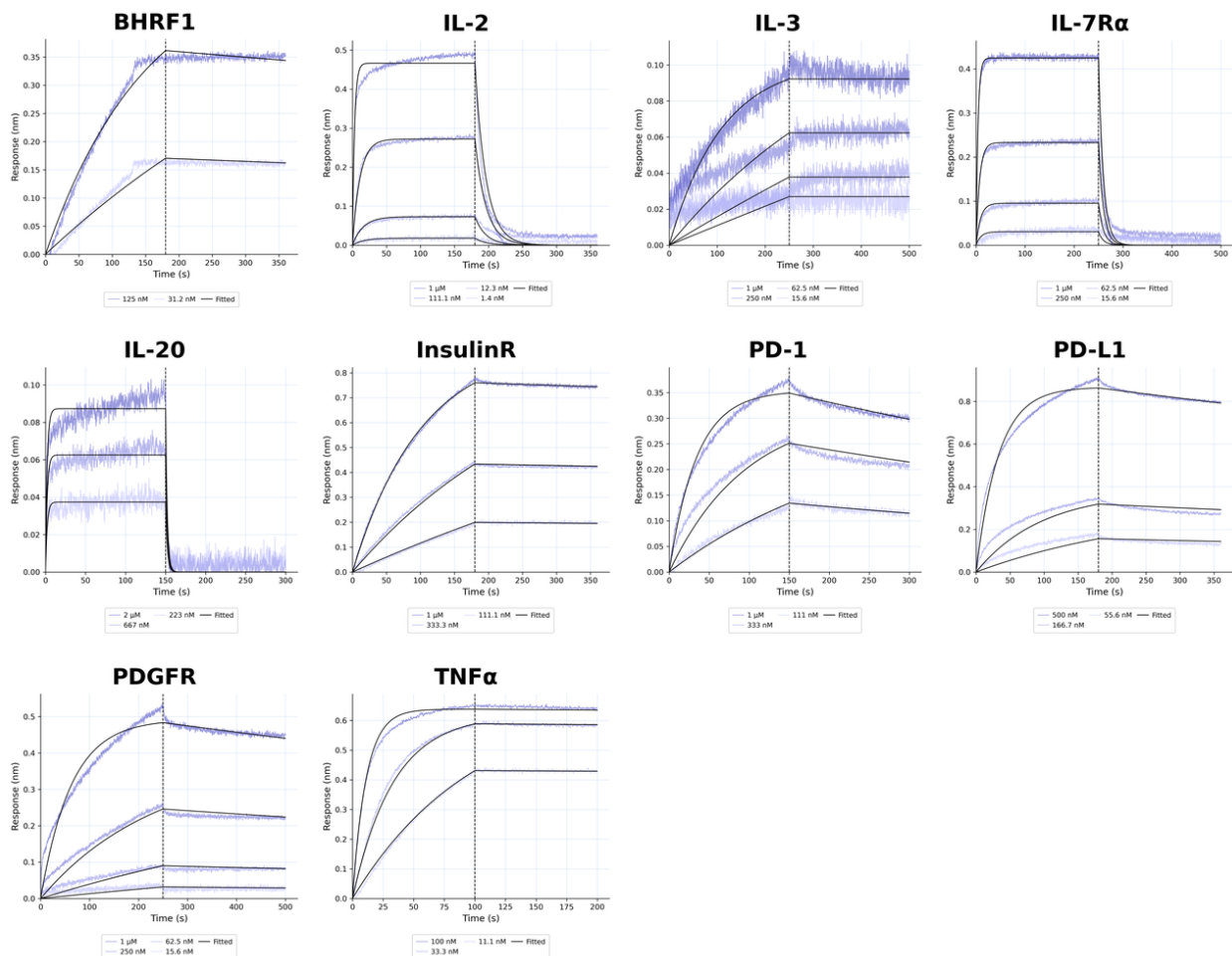


Figure 2: Experimental validation of computationally prioritized VHH candidates. For each target, the predicted VHH–antigen complex structure (left) and multi-concentration BLI sensorgrams with 1:1 global fit (right) are shown.

2.4 Multi-Tier Computational filtering Enriches Experimental Hit Discovery

A post-hoc analysis of the computational filtering pipeline revealed that no single metric alone was sufficient to robustly identify the highest-quality binders across targets. In particular, structure-prediction confidence metrics such as ipTM and pLDDT showed limited correlation with experimentally measured affinity among confirmed binders, consistent with previous observations that confidence metrics may function primarily as coarse binary indicators of binding compatibility rather than quantitative affinity predictors.

Instead, the layered integration of multiple filtering criteria consistently produced stronger enrichment than any individual metric in isolation. The filtering cascade incorporated four major components: (i) structural confidence evaluation, (ii) reproducibility analysis across random seeds and prediction models, (iii) protein language model–based sequence naturalness assessment, and (iv) in-house statistical physics–aware interface scoring. Designs passing all filtering stages were substantially enriched for experimentally confirmed binders relative to candidates selected using any single criterion alone.

These observations suggest that the central challenge in practical AI-native biologics discovery

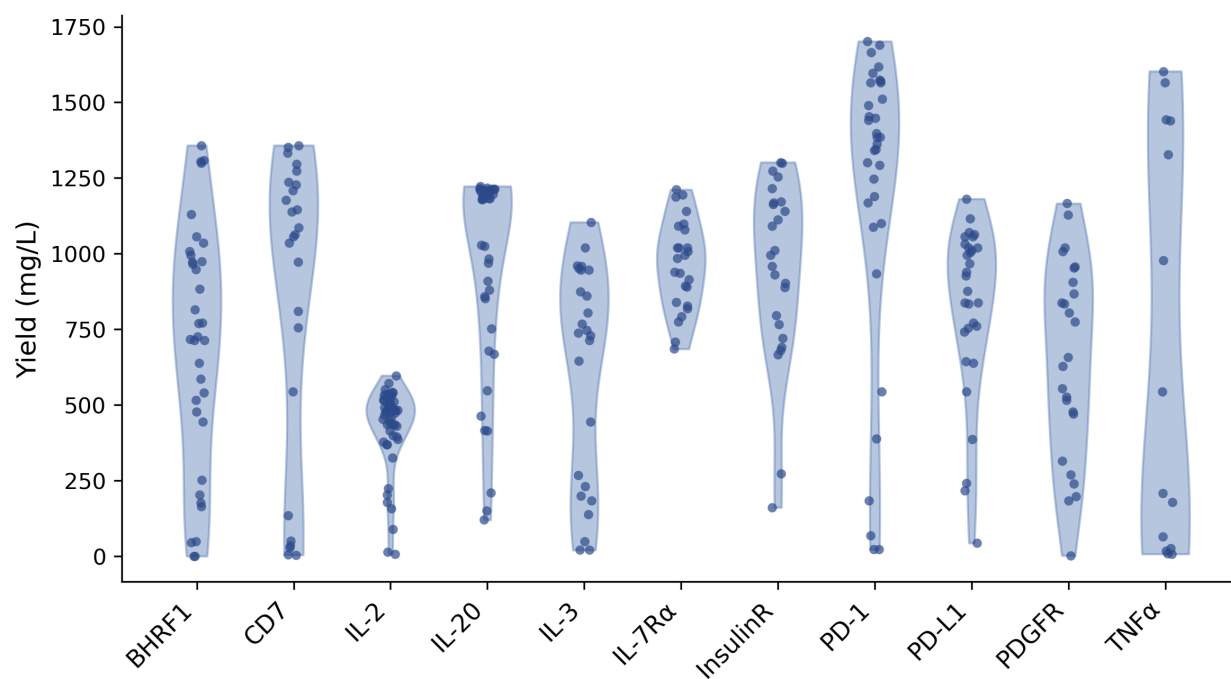


Figure 3: Expression yields (mg/L) of VHH-Fc constructs across all target campaigns. Each data point represents one designed candidate expressed in CHO at 10 mL scale.

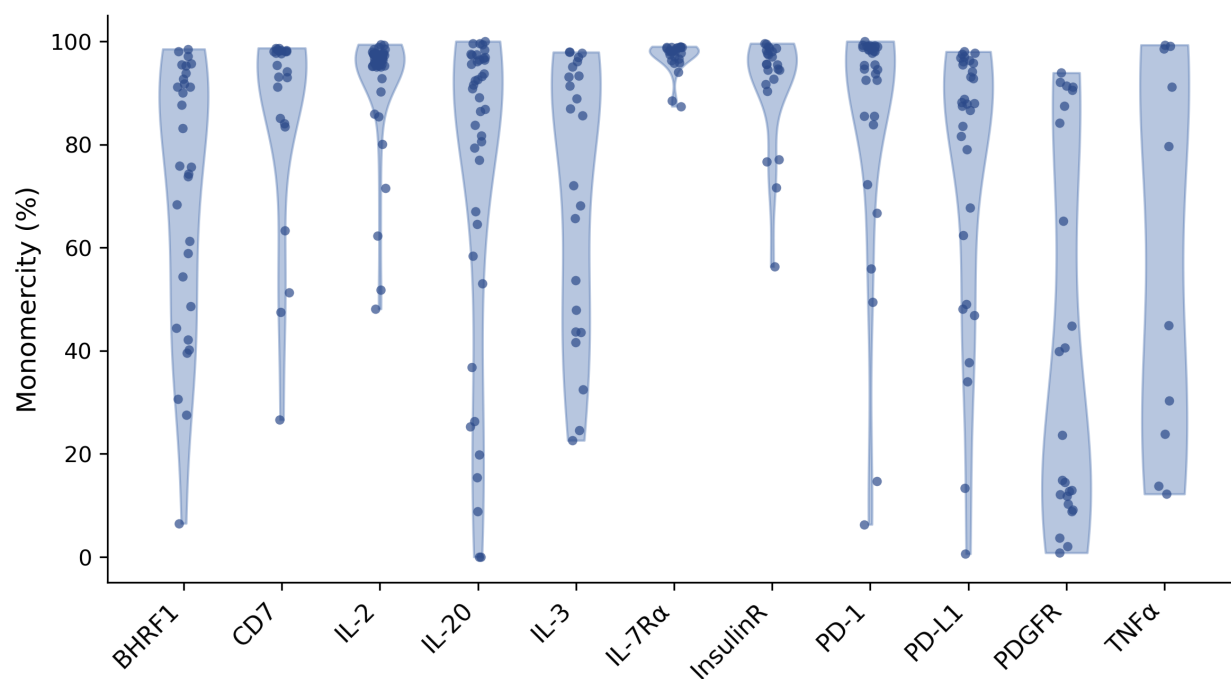


Figure 4: SEC-HPLC chromatograms of representative VHH-Fc candidates. The dominant peak indicates the monomeric VHH-Fc species; minor peaks, if present, correspond to aggregates or degradation products.

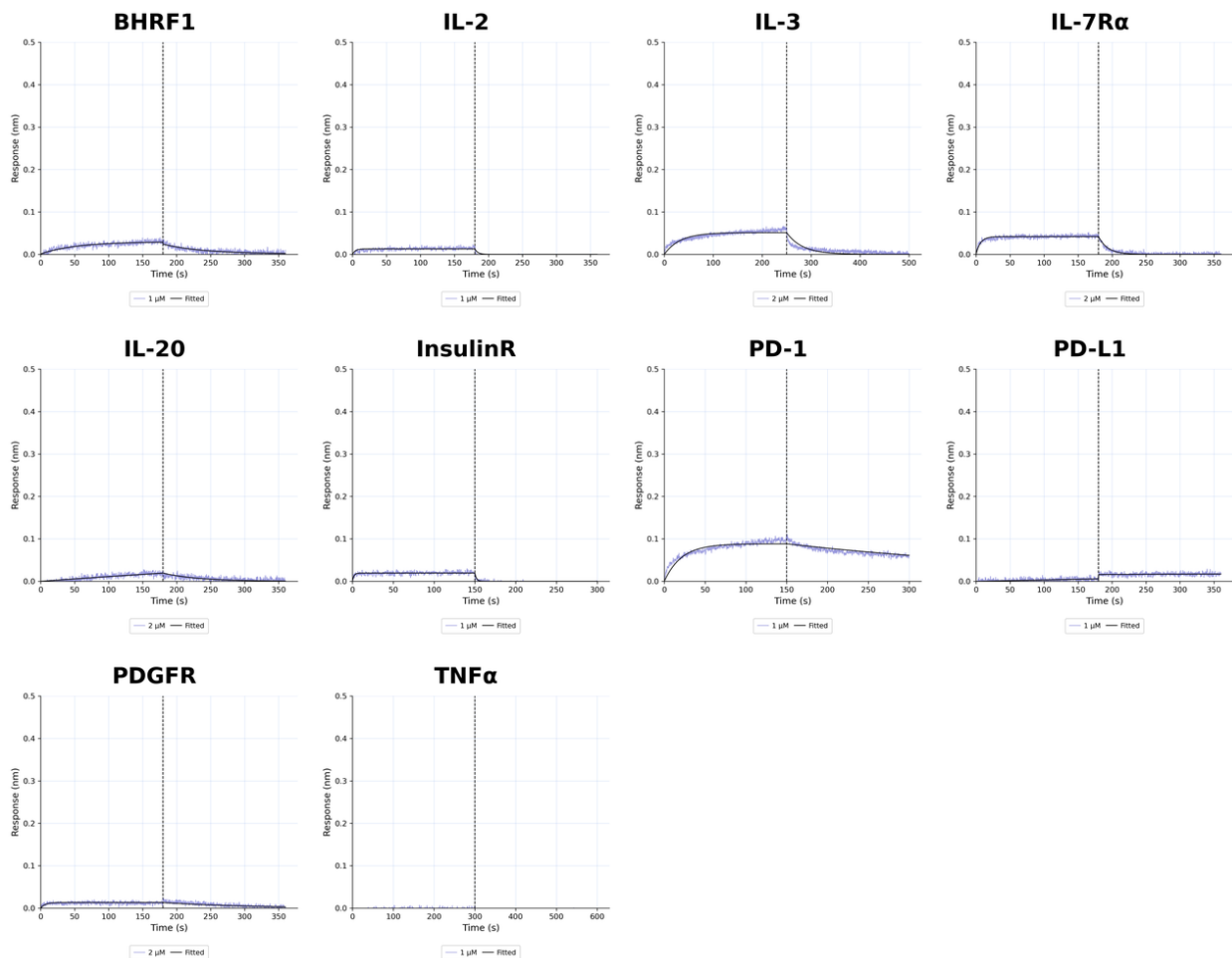


Figure 5: BLI sensorgrams of seven designed VHH candidates against HSA at 1–2 μ M. The dashed vertical line marks the transition from association to dissociation. Blue: experimental response; black: 1:1 kinetic fit.

is not merely generating candidate sequences, but enriching true binders sufficiently to minimize experimental burden. In this context, multi-objective computational prioritization appears to play a critical role in enabling experimentally practical low-throughput discovery workflows.

2.5 MMFold Achieves High Accuracy on Antibody–Antigen Complex Prediction

Because hallucination-guided optimization depends on the quality of the underlying structure-prediction confidence landscape, we evaluated MMFold on the FoldBench [8] antibody–antigen benchmark dataset. MMFold is a proprietary all-atom structure prediction model inspired by the AlphaFold 3 architecture [9] and trained with enhanced antibody–antigen modeling capability. MMFold was trained on Protein Data Bank [10] entries released before September 30, 2021.

Across 172 antibody–antigen interfaces, MMFold achieved an overall DockQ [11] acceptable-or-better success rate of 68.6% for Top-1 predictions, compared with 47.9% for AlphaFold 3 on the same benchmark. Improvements were also observed across medium- and high-quality DockQ thresholds. When expanded to Top-5 predictions, MMFold further increased acceptable-or-better performance to 75.6% (Figure 6). The accuracy of MMFold also scales well along with the number

of sampled predictions. For example, when a few hundred seeds are used, MMFold may achieve high resolution success rate of around 50% for Top-5 predictions.

Several recently released structure-prediction models were not included in this comparison. Boltz-2 [12] was excluded because its more recent training cutoff overlaps with part of the FoldBench release window, which precludes a leakage-free evaluation on this benchmark; notably, the authors themselves report that “Boltz-2 shows an improvement over Boltz-1 while still lagging behind AlphaFold3.” Protenix-v2 [13] reports a DockQ acceptable-or-better success rate of 65% for Top-1 predictions, but this figure was obtained on a 103-PDB-ID (160-interface) subset of the FoldBench antibody–antigen benchmark rather than the full 113-PDB-ID (172-interface) set used here. It is therefore not directly comparable to the results in Figure 6 and is not included.

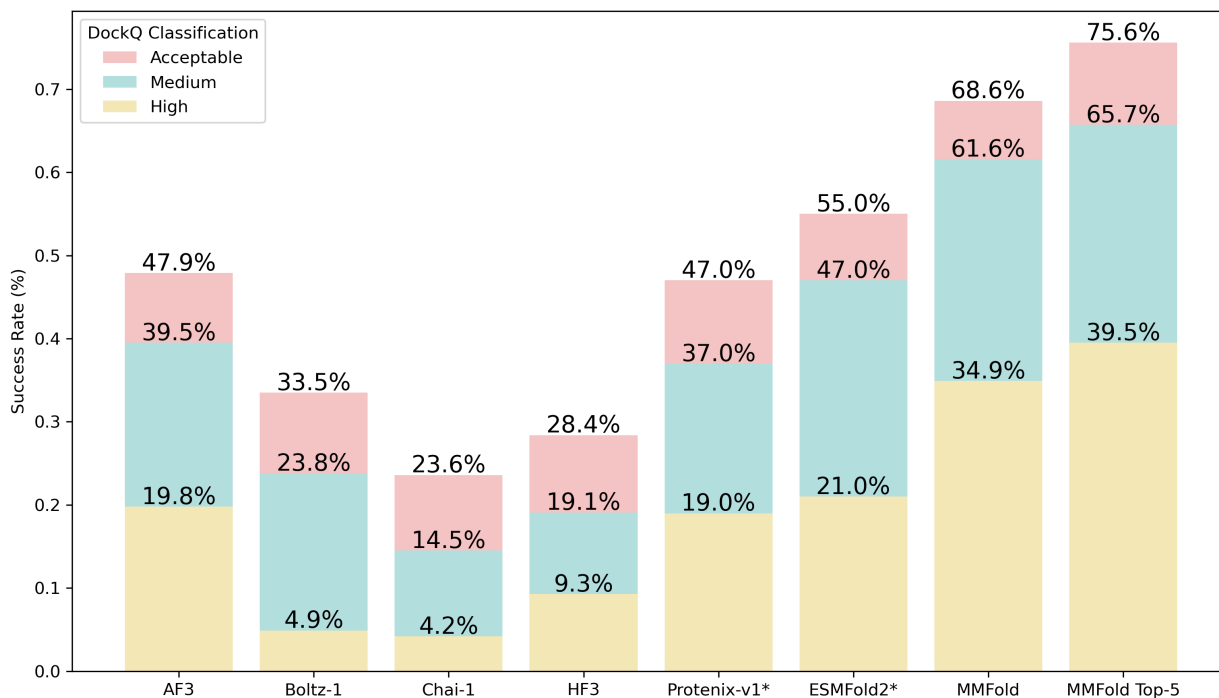


Figure 6: Performance comparison of MMFold against leading structural prediction models based on DockQ classification. The stacked bar chart illustrates the cumulative success rates (%) at High (yellow), Medium (teal), and Acceptable (pink) quality thresholds. MMFold demonstrates significantly higher Top-1 and Top-5 success rates across all quality tiers compared to baselines including AlphaFold 3, Boltz-1, Chai-1, HelixFold3, Protenix-v1 [14], and ESMFold2 [15]. Values marked with an asterisk (Protenix-v1* and ESMFold2*) are reported and inferred from the ESMFold2 paper [15] on the FoldBench antibody–antigen benchmark; the ESMFold2 result corresponds to its 20-loop inference setting. All other models were evaluated here under a single, consistent protocol.

Representative case studies demonstrated substantially improved interface modeling relative to AlphaFold 3 for multiple antibody–antigen complexes, including examples in which MMFold accurately recovered binding geometries that were poorly modeled by baseline systems (Figure 7). Detailed benchmark procedures and additional benchmarking analyses are provided in the Supplementary Information.

Although the current study does not directly establish a causal relationship between structure-prediction accuracy and experimental discovery success, these results suggest that improved

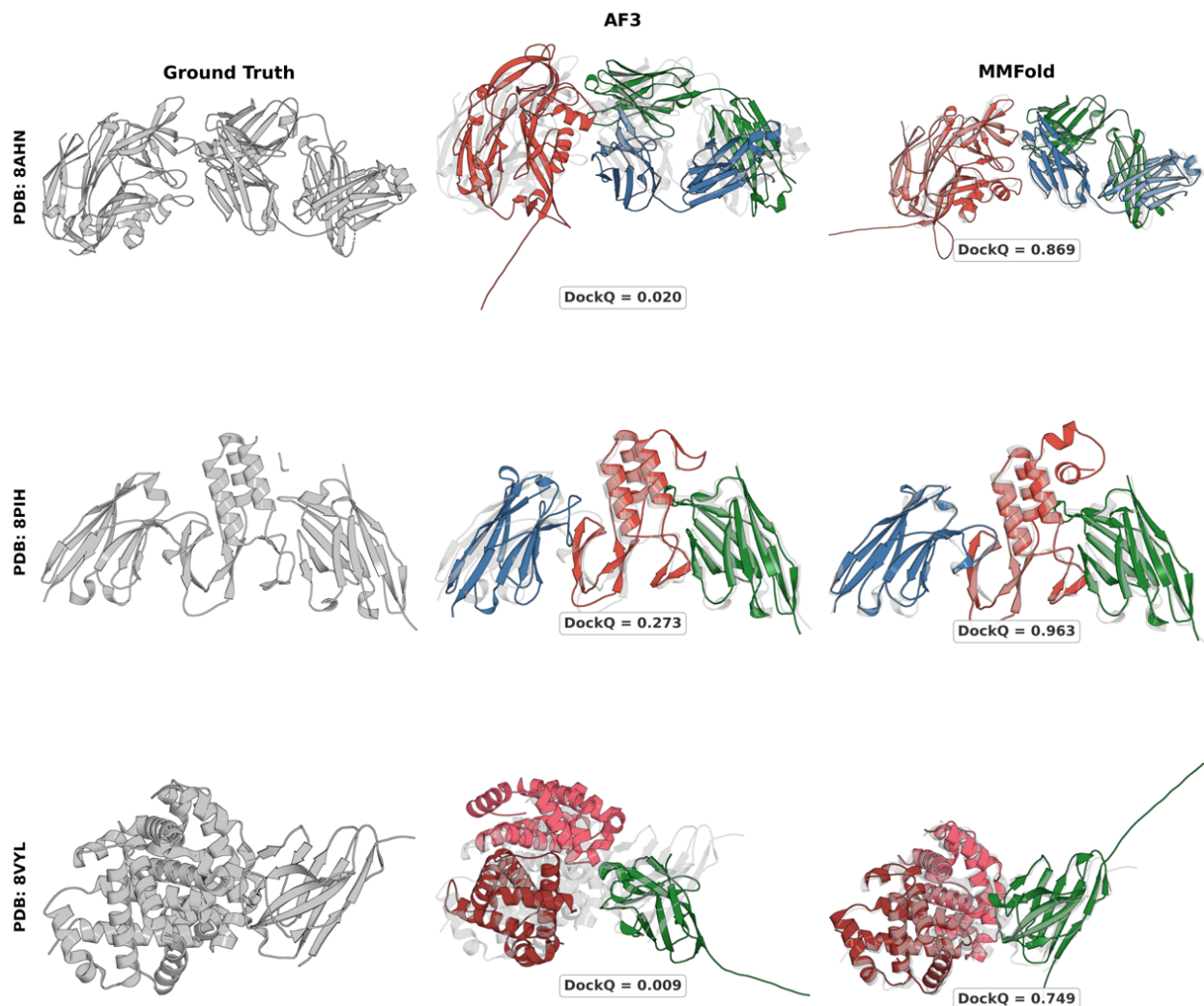


Figure 7: Case studies illustrating the prediction accuracy of antibody–antigen protein complexes. The experimental ground truth (left column) is depicted in solid gray. In the prediction panels (middle: AlphaFold 3; right: MMFold), predicted models are color-coded by subunit—with antigens in red—and superimposed onto the translucent ground truth to visualize structural deviations.

antibody-antigen structure modeling may provide higher-quality confidence landscapes for downstream generative biologics optimization workflows.

3 Discussion

Recent advances in generative modeling have established the possibility of *de novo* antibody and nanobody generation. The results presented here suggest that the field may now be beginning to transition from proof-of-concept demonstrations toward experimentally practical discovery workflows. Across 11 therapeutically relevant targets, MMDesign produced confirmed binders for 10 targets while experimentally testing only 14–50 candidates per campaign, spanning cytokines, immune checkpoints, receptors, a viral protein, and a surface antigen. These findings suggest that generative VHH discovery can increasingly operate at experimentally tractable scales rather than

relying on massive stochastic screening campaigns.

A defining feature of practical AI-native biologics discovery may ultimately be the ability to dramatically compress experimental screening burden while maintaining broad target-level success. Conventional antibody discovery workflows, including animal immunization and display-library screening, typically require exploration of millions to billions of variants and provide limited control over epitope specificity. In contrast, MMDesign starts from only a target protein and user-specified epitope residues, computationally generates tens of thousands of candidate VHHs, and narrows this space to only tens of experimental designs through layered computational enrichment. More broadly, these results suggest a possible transition in biologics discovery from stochastic experimental screening toward programmable molecular engineering.

Sequence novelty of designed binders. A key consideration for any generative design platform is whether the output sequences represent genuinely novel molecules or merely recapitulate known antibodies. The MMDesign generative core is a general-purpose all-atom structure-prediction model that was not specifically fine-tuned on antibody–antigen complexes; the hallucination-based optimization explores the model’s confidence landscape without reference to any antibody sequence database. Consequently, the generation process cannot, by construction, copy or derive sequences from existing binders—whether deposited in public databases or disclosed in patent filings. CDR similarity analysis (Appendix A) and structural comparison against SAbDab (Appendix B) confirm that the designed sequences are substantially divergent from known antibodies, with framework regions preserving canonical VHH scaffold geometry (per-region C α -RMSD consistently below 0.7 Å) while CDR loops—particularly CDR3—adopt conformations largely absent from the experimental database. Binding-pose comparison against known PDB complexes (Appendix C) provides further evidence of structural novelty: 9 of the 11 targets have no nanobody complex in the PDB, so the designed VHHs occupy previously uncharacterized structural space. Where validated references do exist, the designs independently converge on the same functional epitopes despite entirely *de novo* CDRs—the TNF α design recovers the 5M2J epitope (centroid distance 1.03 Å), and the PD-1 design lands on the natural PD-1:PD-L1 interface (centroid distance 0.79 Å, 14 of 24 contacts shared), consistent with a competitive-inhibition mechanism.

The TNF α campaign is particularly notable in this context. TNF α is a homotrimeric cytokine with a relatively shallow and solvent-exposed binding surface that has historically proven challenging for low-throughput *de novo* VHH discovery efforts. In this study, experimental testing of only 14 candidates yielded 7 confirmed binders, corresponding to a 50% hit rate, with the lead molecule reaching an apparent K_D of 51 pM in the avidity-enhanced trimeric setting. Although further characterization will be required to evaluate the generality of this result, it suggests that hallucination-guided generation may begin to access difficult antigen geometries that have historically resisted computational low-throughput discovery approaches.

An additional notable aspect of the current study is that many designed VHHs exhibited favorable preliminary developability-related properties beyond target binding alone. Across campaigns, numerous candidates achieved robust recombinant expression in CHO cells, remained predominantly monomeric by SEC-HPLC, and showed minimal detectable cross-reactivity against HSA in BLI assays. Although these measurements do not constitute a comprehensive developability assessment, they suggest that generative discovery systems may increasingly be capable of producing experimentally tractable antibody-like molecules rather than merely isolated binding sequences. More broadly, these observations raise the possibility that future generative antibody platforms may optimize not only for binding activity, but simultaneously for expression, stability, specificity, and downstream therapeutic developability.

The current study also highlights the potential importance of integrating generation and compu-

tational filtering within a unified workflow. Post-hoc analysis indicated that no individual metric alone—including structure-prediction confidence measures such as ipTM or pLDDT—was sufficient to robustly identify the highest-quality binders across targets. Instead, the layered combination of structural reproducibility analysis, sequence naturalness evaluation, epitope-contact verification, and physics-based interface assessment consistently provided stronger enrichment than any single criterion in isolation. These observations suggest that practical generative antibody discovery may depend not only on generative capability itself, but also on effective multi-objective prioritization strategies capable of filtering large computational candidate spaces into experimentally actionable sets.

Several limitations remain. First, although confirmed binding was demonstrated across diverse targets, affinity remains variable across campaigns, with many successful binders clustering in the nanomolar regime rather than consistently achieving sub-nanomolar potency. Improving affinity consistency across diverse antigen classes remains a major objective for future model development. Second, the current study primarily relies on predicted complex structures and binding assays; experimental structural validation of designed antibody–antigen interfaces will be important for establishing the accuracy of the generated binding geometries and understanding the mechanisms underlying successful designs. Third, while preliminary expression and specificity measurements are encouraging, comprehensive therapeutic developability assessment—including long-term stability, aggregation resistance, immunogenicity, and *in vivo* pharmacokinetic profiling—was beyond the scope of the present work.

Several directions are under active development. Continued improvements in antibody–antigen structure modeling and generative optimization may enable the platform to address increasingly difficult target classes, including multispecific biologics, GPCRs and other membrane proteins. Beyond *de novo* binder generation, preliminary internal results suggest that MMDesign may be extendable to more advanced antibody engineering settings, including context- or pH-dependent binding optimization. In parallel, expanding the design–experiment feedback loop may progressively transform antibody discovery into a self-improving closed-loop optimization process in which experimental outcomes continuously refine generative and filtering performance. Finally, incorporation of broader developability-aware objectives may allow future systems to optimize not only for binding, but also for manufacturability and therapeutic fitness from the earliest stages of molecular generation.

Taken together, these results suggest that *de novo* nanobody generation may increasingly evolve from an exploratory research paradigm into a practical therapeutic discovery modality. If continued improvements in generation, filtering, and experimental feedback can be sustained, computationally guided antibody discovery may ultimately transition from large-scale stochastic screening toward programmable molecular engineering.

Availability

The MMDesign platform is available for collaborative target campaigns with pharmaceutical and biotechnology partners. MMFold is available as a feature of MoleculeOS at mos.moleculemind.com.

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A CDR Similarity Analysis of Designed VHH Sequences Against Known Antibody Databases

A.1 Methods

To assess the novelty of designed sequences relative to known antibody sequences, we performed a CDR similarity analysis on 338 VHH sequences across eleven target systems, and additionally examined the highest-affinity VHH candidate from each system.

Designed sequences were first searched against a local PDB protein sequence database (1,043,893 protein chains) using blastp (BLAST+ v2.15.0, E-value $< 10^{-5}$, top 5 hits per query). For hits from the same PDB entry, only the chain with the highest identity was retained. Both query and hit sequences were then numbered using the IMGT scheme via ANARCI. Residue-level differences were counted within CDR1 (IMGT 27–38), CDR2 (IMGT 56–65), and CDR3 (IMGT 105–117) by aligning positions based on their IMGT numbers and recording mismatches, insertions, and deletions. For each designed sequence, the PDB hit with the highest overall sequence identity was selected as the comparison reference.

To further assess novelty beyond structurally characterized proteins, the highest-affinity VHH candidate from ten of the eleven target systems (excluding CD7, for which no experimentally confirmed binder was identified) was additionally searched against the NCBI non-redundant protein database (nr, ~250 million sequences) using blastp (E-value $< 10^{-5}$, top 20 hits). The entry with the highest sequence identity was selected, and CDR mutations were computed using the same IMGT-based method. Top hits were manually reviewed to determine whether any known VHH or nanobody targeting the same antigen was present in the database.

To assess sequence novelty against the broader space of naturally observed antibody repertoires, all 338 designed VHH sequences were additionally searched against the Observed Antibody Space (OAS) heavy-chain database using jackhmmer (HMMER 3.3, 1 iteration, $E < 10^{-4}$, up to 9,999 hits per query). OAS is a curated repository of antibody sequences derived from high-throughput B-cell receptor sequencing studies. Because VHH domains are structurally equivalent to the heavy-chain variable region, only the heavy-chain sub-database was used for this comparison; light-chain entries were excluded. Identity between a designed sequence and each OAS hit was computed as the fraction of aligned positions (excluding insertions relative to the query) that match the query residue. Per-sequence statistics were then aggregated across all designed sequences within each target system.

A.2 Results

CDR mutation statistics across all designed sequences are summarized in Table 2, and the highest-affinity VHH candidate per system is shown in Table 3 (CD7 is excluded from Tables 3 and 4 as no VHH candidate with experimentally confirmed binding affinity was identified for this target).

Results of the NCBI nr search for the highest-affinity VHH candidate per system (excluding CD7) are shown in Table 4.

The PD-L1 binder showed the highest homology to a known structure in the PDB (anti-PD-L1 VHH, PDB: 7czd), with a sequence identity of 92.2% and only 9 CDR mutations. For the remaining ten systems, best-hit identities were below 81% and CDR total mutations of the highest-affinity candidates ranged from 7 to 31, consistent with the population-level statistics in Table 2. Across all systems, CDR3 consistently harbored the largest number of mutations, consistent with its role as the primary determinant of antigen-binding diversity. Searching the broader NCBI nr database confirmed that six of eleven targets (IL-2, IL-3, IL-20, PDGFR, BHRF1, and InsulinR) have no previously reported VHH or nanobody in the public sequence databases, indicating that the designed candidates for these targets are not direct derivatives of previously disclosed sequences.

To quantify novelty at the per-region level, each designed sequence was aligned against the OAS heavy-chain hit with the lowest E-value (i.e., the closest sequence match among 9,999 retrieved sequences), and mutations were counted separately for each IMGT-numbered CDR and FR region (annotated via abnumber). Per-region statistics are presented in Table 5 (all designed sequences per system) and Table 6 (highest-affinity candidate per system, excluding CD7).

Table 2: CDR mutation statistics of all designed sequences versus their best PDB hits (IMGT numbering).

System	N	Best PDB hit	CDR1		CDR2		CDR3		CDR
		Mean identity (%)	Mean	Range	Mean	Range	Mean	Range	Total mean
PD-L1	30	92.2	0.4	0–1	2.4	1–5	6.1	2–8	8.9
TNF α	14	76.4	1.5	0–5	2.4	1–4	6.4	5–8	10.3
PD-1	44	75.8	3.2	1–6	4.5	2–6	11.1	7–18	18.9
PDGFR	25	79.9	5.3	4–7	5.6	5–6	13.1	11–20	24.1
IL-7R α	25	79.8	6.5	5–7	4.8	2–7	13.6	11–15	24.9
CD7	25	78.5	5.1	2–7	4.0	1–6	17.9	16–21	27.0
BHRF1	35	78.4	8.5	6–11	4.3	2–8	14.5	11–17	27.3
IL-20	40	78.2	8.6	5–11	4.1	1–6	15.0	11–18	27.7
InsulinR	25	78.5	4.5	4–8	4.5	2–7	18.7	16–21	27.8
IL-3	25	77.5	8.5	5–10	5.4	4–7	14.9	12–17	28.8
IL-2	50	77.0	9.8	7–11	5.2	3–8	15.2	11–18	30.2

N: number of designed sequences. Best PDB hit mean identity: blastp sequence identity over the full VHH domain sequence. CDR1/CDR2/CDR3 Mean and Range: mean and min–max number of mutated residues in each CDR region compared with the best PDB hit. CDR Total mean: mean of the sum of CDR1+CDR2+CDR3 mutations. Systems are ordered by CDR total mean in ascending order.

Table 3: CDR mutation counts of the highest-affinity VHH candidate per system versus its best PDB hit (IMGT numbering).

System	Best PDB hit	Mutations vs. best PDB hit				
	Identity (%)	CDR1	CDR2	CDR3	FR	CDR Total
TNF α	79.1	0	1	6	17	7
PD-L1	92.2	0	2	7	0	9
PD-1	76.7	6	4	10	9	20
IL-7R α	79.3	7	4	14	0	25
IL-20	78.9	8	5	14	0	27
IL-3	77.3	7	5	17	0	29
BHRF1	77.7	10	6	13	0	29
IL-2	76.0	11	5	14	1	30
InsulinR	77.5	5	6	19	1	30
PDGFR	76.6	6	5	20	0	31

The highest-affinity VHH candidate is shown per system. Best PDB hit identity: blastp sequence identity over the full VHH domain sequence. CDR1/CDR2/CDR3/FR: number of mutated residues in each region compared with the best PDB hit (IMGT numbering). CDR Total: sum of CDR1+CDR2+CDR3 mutations. Systems are ordered by CDR total in ascending order.

Table 4: CDR mutation counts of the highest-affinity VHH candidate per system versus its closest match in NCBI nr (IMGT numbering).

System	Best nr hit Identity (%)	Mutations vs. best nr hit				
		CDR1	CDR2	CDR3	FR	CDR Total
TNF α	76.5	1	1	6	19	8
PD-L1	92.2	0	2	7	0	9
PD-1	76.7	6	4	10	9	20
IL-7R α	79.3	7	4	14	0	25
PDGFR	78.5	6	6	13	1	25
IL-20	78.9	8	5	14	0	27
BHRF1	78.9	8	5	14	0	27
IL-3	78.1	8	6	15	0	29
IL-2	77.0	10	6	13	0	29
InsulinR	77.5	5	6	19	1	30

The highest-affinity VHH candidate is shown per system. Best nr hit identity: blastp sequence identity over the full VHH domain against NCBI nr (~250 million sequences, E-value < 10^{-5} , top 20 hits), selecting the entry with the highest identity. CDR1/CDR2/CDR3/FR: number of mutated residues in each region (IMGT numbering). CDR Total: sum of CDR1+CDR2+CDR3 mutations. Note that best nr hits are the closest sequences by identity and do not necessarily target the same antigen. Systems are ordered by CDR total in ascending order.

Table 5: CDR and FR mutation statistics of all designed VHH sequences versus their best OAS heavy-chain hit (IMGT numbering). FR2 consistently shows 4–9 mutations across all systems owing to VHH hallmark substitutions at positions 37/44/45/47 that differ from human VH sequences in OAS. FR3 of TNF α is a notable outlier (mean 13.7 mutations) reflecting a non-canonical scaffold in those designs. Systems are ordered by CDR Total mean in ascending order.

System	N	CDR1		CDR2		CDR3		CDR Total	FR mutations (mean)				FR Total
		Mean	Range	Mean	Range	Mean	Range		FR1	FR2	FR3	FR4	
PD-L1	30	0.4	0–1	1.4	0–3	3.4	1–6	5.3	1.2	3.9	4.1	2.0	11.1
TNF α	14	1.6	0–4	3.8	3–5	4.6	2–7	10.1	2.2	4.9	13.7	2.1	23.0
PD-1	44	3.8	2–5	2.6	0–6	6.0	3–8	12.3	0.6	8.0	5.4	1.0	15.0
PDGFR	25	3.5	2–6	3.7	3–5	6.8	5–9	14.0	0.2	5.0	5.9	1.0	12.1
IL-7R α	25	4.0	2–7	4.4	1–7	8.0	6–10	16.4	0.5	5.9	3.1	0.0	9.5
InsulinR	25	2.6	0–6	2.9	0–5	13.2	11–18	18.6	0.6	5.1	6.0	1.0	12.6
BHRF1	35	6.9	4–10	3.6	1–6	9.5	6–13	20.0	1.8	6.1	0.1	0.1	8.0
CD7	25	4.0	1–7	3.2	1–5	13.5	10–16	20.6	2.0	8.1	4.4	1.0	15.5
IL-20	40	7.1	4–9	3.7	1–6	10.2	7–15	21.1	1.8	6.0	0.0	0.1	7.9
IL-3	25	7.2	4–10	4.9	3–7	9.6	6–12	21.8	1.9	6.0	0.0	0.1	8.0
IL-2	50	8.6	6–11	4.8	2–8	11.3	8–15	24.7	1.5	6.1	0.3	0.2	8.1

N: number of designed sequences. CDR/FR Mean and Range: mean and min–max mutations per region versus the best OAS heavy-chain hit (IMGT). CDR Total / FR Total: sum of CDR1+CDR2+CDR3 and FR1+FR2+FR3+FR4 mutations, respectively. FR2 mutations reflect VHH-specific hallmark residues (positions 37/44/45/47) absent in human VH sequences (OAS); FR3 mutations indicate scaffold divergence from the natural heavy-chain repertoire.

Table 6: CDR and FR mutation counts of the highest-affinity VHH candidate per system versus its best OAS heavy-chain hit (IMGT numbering). CD7 is excluded as no candidate with experimentally confirmed binding was identified. FR2 mutations (4–9) are present in all systems due to VHH hallmark positions differing from human VH. TNF α shows 14 additional FR3 mutations from a non-canonical scaffold. Systems are ordered by CDR Total in ascending order.

System	CDR mutations				FR mutations				
	CDR1	CDR2	CDR3	CDR Total	FR1	FR2	FR3	FR4	FR Total
PD-L1	0	1	4	5	1	4	4	2	11
TNF α	0	5	5	10	2	5	14	2	23
IL-7R α	2	4	7	13	0	6	3	0	9
PDGFR	4	4	5	13	1	5	6	1	13
PD-1	4	3	7	14	0	9	5	1	15
IL-20	5	5	7	17	2	6	0	0	8
BHRF1	7	4	9	20	1	6	0	0	7
InsulinR	1	5	14	20	1	5	6	1	13
IL-2	10	4	9	23	2	6	0	0	8
IL-3	7	6	12	25	2	6	0	0	8

CDR1/CDR2/CDR3: mutations in each CDR region versus the best OAS heavy-chain hit (IMGT). CDR Total: CDR1+CDR2+CDR3. FR1–FR4: mutations in each framework sub-region. FR Total: FR1+FR2+FR3+FR4.

B Structural Similarity of Experimentally Validated VHH Designs to SAbDab

MMfold-predicted VHH structures (363 designs across 11 target systems) were compared against 3,313 SAbDab experimental antibody structures using pairwise C α -RMSD. For each design, global Needleman–Wunsch sequence alignment was performed, followed by a single Kabsch rigid-body superposition on all aligned C α atoms. Per-region RMSD (IMGT scheme: FR1–FR4, CDR1–CDR3) was then computed under the fixed superposition without re-aligning. The closest SAbDab neighbour for each design was identified by minimum overall RMSD. Experimentally validated VHH sequences (confirmed binders and additional wet-lab-characterised candidates) were matched to MMfold prediction records by exact sequence identity.

Table 7 shows, for each experimentally validated VHH, the overall and per-region RMSD against its single closest SAbDab structure. **Table 8** gives the system-level mean over all designs within each system, serving as a population reference. Framework regions (FR1–FR4) are consistently below 0.7 Å across all systems, confirming canonical VHH scaffold geometry. CDR3 dominates structural novelty: values above 3 Å in PDGFR, InsulinR, and IL-3 indicate loop conformations largely absent from the experimental database, whereas TNF α , PD-L1, and PD-1 (<2 Å) align closely with known antibody loops. The experimentally validated designs match or improve upon their system mean in most cases; PD-L1 is the exception (0.61 vs. mean 0.48 Å) because the entire PD-L1 system already converges near SAbDab.

These analyses indicate that MMDesign does not recover previously known nanobody sequences or simple variants of known binders. Instead, the platform generates VHH-like but sequence-divergent CDR architectures, including target-specific binders with no close known counterpart in public antibody, structural, or patent databases.

Table 7: Per-region RMSD (Å) of each experimentally validated VHH design vs. its closest SAbDab neighbour (minimum overall RMSD). CDR columns in **bold**.

System	Overall	FR1	CDR1	FR2	CDR2	FR3	CDR3	FR4
IL-2	1.310	0.418	3.350	0.309	2.369	0.419	3.087	0.320
PD-L1	0.608	0.264	0.380	0.246	0.475	0.267	2.041	0.249
TNF α	0.537	0.417	0.182	0.424	0.477	0.320	1.482	0.456
IL-7R α	0.782	0.158	0.412	0.195	0.550	0.228	2.696	1.026
IL-3	1.270	0.411	2.959	0.587	1.340	0.380	4.044	1.111
IL-20	1.469	0.426	2.484	0.671	2.379	0.516	3.101	0.924
PDGFR	1.050	0.352	0.915	0.660	1.083	0.342	4.484	1.001
BHRF1	1.229	0.480	2.646	0.533	2.846	0.363	1.526	1.045
InsulinR	1.134	0.231	0.460	0.632	0.454	0.315	4.051	0.347
PD-1	0.891	1.165	1.782	0.667	0.518	0.315	1.453	0.154

Table 8: Mean per-region RMSD (Å) across all designs in each system, each evaluated against its closest SAbDab neighbour. CDR columns in **bold**.

System	Overall	FR1	CDR1	FR2	CDR2	FR3	CDR3	FR4
BHRF1	1.349	0.611	3.009	0.514	1.654	0.426	2.807	0.673
CD7	1.070	0.404	1.304	0.490	1.402	0.350	2.628	0.690
IL-2	1.501	0.600	3.910	0.596	2.135	0.466	3.071	0.671
IL-20	1.369	0.527	3.170	0.545	1.961	0.499	2.756	0.797
IL-3	1.345	0.509	3.176	0.507	1.906	0.475	2.596	0.738
IL-7R α	0.987	0.424	1.775	0.432	1.325	0.341	2.528	0.657
InsulinR	1.072	0.418	1.124	0.480	1.201	0.366	3.151	0.579
PD-1	0.925	0.520	1.380	0.413	1.395	0.312	2.000	0.591
PDGFR	0.949	0.529	1.476	0.502	1.703	0.382	2.042	0.511
PD-L1	0.475	0.182	0.242	0.244	0.436	0.164	1.605	0.324
TNF α	0.483	0.285	0.344	0.419	0.462	0.354	1.175	0.352

C Structural Novelty of Designed VHHs: Binding Pose Comparison Against Known PDB Complexes

VHH sequences were designed *de novo* against 11 targets, of which **9 targets have no existing VHH complex structure in the PDB** (BHRF1, CD7, IL2, IL20, IL3, IL7R α , InsulinR, PD1, PDGFR); only PD-L1 and TNF α have known VHH complexes available for comparison. This analysis provides quantitative binding-pose comparisons to demonstrate: (1) for targets lacking VHH references, the designed VHHs occupy previously uncharacterised structural space while targeting functionally relevant epitopes; (2) for targets with known VHH structures, the designed VHHs display identical binding modes to the reference VHHs despite entirely *de novo* CDR sequences, confirming design validity.

C.1 Methods

Reference structure retrieval For each target, known complex structures were retrieved from the RCSB PDB via sequence-similarity search (identity threshold 80%) using the target protein sequence as query, and cross-annotated against the SAbDab database to classify entries as VHH complexes, IgG/Fab complexes, or

natural protein–protein complexes.

Step 1: Antigen chain identification Each protein chain in a reference structure was aligned against the target sequence using local pairwise alignment (Biopython PairwiseAligner, local mode); the alignment score normalised by query length was used as the identity measure, and the highest-scoring chain was designated the antigen chain (structures with identity < 0.4 were excluded). Local alignment correctly handles cases where the query sequence is a sub-domain of a longer PDB chain (e.g. an extracellular domain of a full-length receptor).

Step 2: Antigen-chain superimposition The antigen chain of each reference structure was superimposed onto chain A (target protein) of the designed complex:

1. Global pairwise sequence alignment was performed between the two antigen chains; gap-free matched positions yielding $C\alpha$ atom pairs were extracted (minimum 10 pairs required).
2. The optimal rotation matrix $\mathbf{R}$ and translation vector $\mathbf{t}$ minimising $C\alpha$ RMSD were computed by singular value decomposition (SVD).
3. The transformation $(\mathbf{R}, \mathbf{t})$ was applied to **all atoms** of the reference structure, preserving the relative geometry of all chains in the complex.

After superimposition, the position of the reference binder chain in the design coordinate frame represents its true binding pose relative to the target.

Step 3: Interface residue identification A 5 Å heavy-atom distance cutoff was used to define interface residues: a residue is classified as an interface residue if any of its heavy atoms lies within 5 Å of any heavy atom of the opposing chain. This yields two sets: E_{design} (target residues contacted by the designed VHH, numbered according to the design structure) and E_{ref} (target residues contacted by the reference binder, numbered according to the reference structure).

Step 4: Binding-pose metrics (1) **Epitope Centroid Distance.** After superimposition, the $C\alpha$ centroids of the two interface residue sets, $\bar{\mathbf{c}}_{\text{design}}$ and $\bar{\mathbf{c}}_{\text{ref}}$, are computed and their Euclidean distance taken:

$$d_{\text{epi}} = \|\bar{\mathbf{c}}_{\text{design}} - \bar{\mathbf{c}}_{\text{ref}}\|$$

A smaller d_{epi} indicates the two binders engage the same spatial location on the target.

(2) **Approach Vector Angle.** For each binder, the approach vector is defined as the unit vector pointing from the epitope centroid to the binder’s centre of mass:

$$\mathbf{v} = \frac{\bar{\mathbf{c}}_{\text{binder}} - \bar{\mathbf{c}}_{\text{epi}}}{\|\bar{\mathbf{c}}_{\text{binder}} - \bar{\mathbf{c}}_{\text{epi}}\|}$$

The angle $\theta = \arccos(\mathbf{v}_{\text{design}} \cdot \mathbf{v}_{\text{ref}})$ measures the directional agreement between the two binders (0: identical approach direction; 180: opposite directions).

(3) **Jaccard Similarity.** Epitope residue overlap is quantified by the Jaccard index. Because residue numbering schemes frequently differ between the design structure and reference structures (offsets up to +35 were observed), direct comparison of residue numbers produces systematic false negatives. Residues in E_{design} and E_{ref} are therefore both mapped to alignment columns derived from the sequence alignment of the two antigen chains before computing:

$$J = \frac{|E_{\text{design}} \cap E_{\text{ref}}|}{|E_{\text{design}} \cup E_{\text{ref}}|}$$

$J \in [0, 1]$; higher values indicate greater overlap in the target residues contacted by the two binders.

C.2 Results

Table 9: Binding-pose metrics for all targets (valid comparisons only). Shaded rows: targets with VHH or natural-ligand references (PD-L1, TNF α , PD1). Green bold: all three metrics indicate high consistency (centroid dist. <3 Å, approach angle $<32^\circ$, Jaccard >0.5). Ab = IgG/Fab complex; VHH = nanobody complex; PD1:PD-L1 = natural protein–protein complex.

Target	Ref. PDB	Type	Centroid dist. (Å)	Approach angle ($^\circ$)	Jaccard	Novelty note
BHRF1	4OYD	Ab	6.80	123.0	0.24	No known VHH; binding mode differs markedly from reference Ab
IL2	8SOZ	Ab	4.48	67.4	0.32	No known VHH; approach angle distinct from known Abs
	8SOW	Ab	7.44	79.9	0.26	
IL20	4DOH	Ab	14.67	115.0	0.07	No known VHH; binding region entirely distinct from reference Ab
IL3	5UWC	Ab	5.91	28.5	0.52	No known VHH; similar direction to 5UWC; others diverge
	5UV8	Ab	6.49	107.0	0.23	
	6NMY	Ab	5.78	103.5	0.34	
IL7Ra	7OPB	Ab	5.27	31.6	0.36	No known VHH; two reference Abs show inconsistent poses
	5J11	Ab	16.05	27.1	0.60	
InsulinR	2HR7	Ab	3.73	73.6	0.18	No known VHH; approach angle 73° off nearest reference Ab
PD1	3BIK	PD1:PD-L1	0.79	19.5	0.56	No known VHH; targets PD1:PD-L1 natural interface; 14/24 contacts shared with PD-L1
PD-L1	7CZD	VHH	0.66	3.6	0.58	Known VHH refs exist; same epitope, <i>de novo</i> CDRs
	8AOM	VHH	1.92	27.6	0.56	
	5JDS	VHH	2.32	15.9	0.60	
	8AOK	VHH	5.27	31.5	0.53	
TNF α	5M2J	VHH	1.03	4.8	0.80	Known VHH refs exist; matches 5M2J epitope
	5M2I	VHH	7.87	122.4	0.17	
	5M2M	VHH	19.49	73.5	0.00	
	21TW	VHH	5.82	101.8	0.29	

C.3 Key Findings

1. **Nine targets have no known VHH complex structure in the PDB.** BHRF1, CD7, IL2, IL20, IL3, IL7Ra, InsulinR, PD1, and PDGFR have no deposited nanobody complex in the public structural database, indicating that the binding modes of the designed VHHs cannot be directly benchmarked against experimental VHH references for these targets.
2. **The designed PD1 VHH substantially overlaps the native PD1:PD-L1 interaction interface.** Comparison against the PD1:PD-L1 co-crystal structure (3BIK) yields a centroid distance of 0.79 Å, an approach angle of 19.5°, and a Jaccard overlap coefficient of 0.56. Residue-level analysis further shows that 14 of 24 VHH contact residues (58%) are shared with the natural PD-L1 binding interface, consistent with a potential competitive-inhibition mechanism. Together, these results suggest that MMDesign independently converged onto a functionally relevant inhibitory interface.
3. **PD-L1 and TNF α designs confirm sequence novelty at a validated functional epitope.**

For both targets with known VHH references, the designed VHHs occupy the same functional epitope as existing VHHs (PD-L1: centroid dist. 0.7–2.3 Å; TNF α /5M2J: 1.03 Å) while bearing entirely *de novo* CDR sequences with no direct homology to the reference VHHs, demonstrating sequence-level novelty.

The designed TNF α VHH differs from the reference TNF α VHH (5M2J) by 10 aligned CDR positions, including two insertion/deletion events, indicating substantial divergence in the antigen-recognition regions rather than simple point substitution of an existing binder. That is, MMDesign can independently recover therapeutically validated binding solutions through *de novo*-generated antigen-recognition sequences, despite no exposure to the patented references during model training.

The designed PD-L1 VHH differs from the reference (7CZD) by 9 CDR positions. The PD-L1 campaign demonstrates that MMDesign can reproducibly recover therapeutically validated epitope geometries through independently generated VHH sequences, supporting the biological realism of the design process.
4. **Several targets display binding modes distinct from the available IgG/Fab reference structures.** The designed VHHs for IL2 (approach angle 67°–80°), IL20 (centroid dist. 14.7 Å), and BHRF1 (approach angle 123°) diverge substantially from the IgG/Fab complexes available in the PDB, indicating that these designs engage the target from orientations not represented among the retrieved reference antibody structures.

D Methods

D.1 Design Overview

The MMDesign pipeline follows a “generate-then-filter” strategy and consists of four stages: target preparation, sequence–structure generation, computational filtering, and experimental characterization. In the generation stage, a structure-prediction model operates in a hallucination paradigm [16] to co-generate VHH CDR sequences and full-atom complex structures from a specified epitope, producing tens of thousands of candidates per target. In the filtering stage, multi-tier computational filters—covering sequence expressibility, structural reliability, and physics-based binding scores—progressively reduce the candidate pool to tens of designs for wet-lab testing. The overall workflow is illustrated in Figure 8.

D.2 Target Preparation

Each campaign requires only an antigen sequence and a set of epitope hotspot residue positions. VHH scaffold template and CDR loop lengths can be user-specified or automatically assigned.

D.3 Sequence–Structure Generation

The generative core operates in a hallucination paradigm: given the antigen sequence, hotspot positions, and VHH scaffold, the CDR sequences are optimized through backpropagation of customized loss functions through both the structure-prediction model and a protein language model. These losses encode design

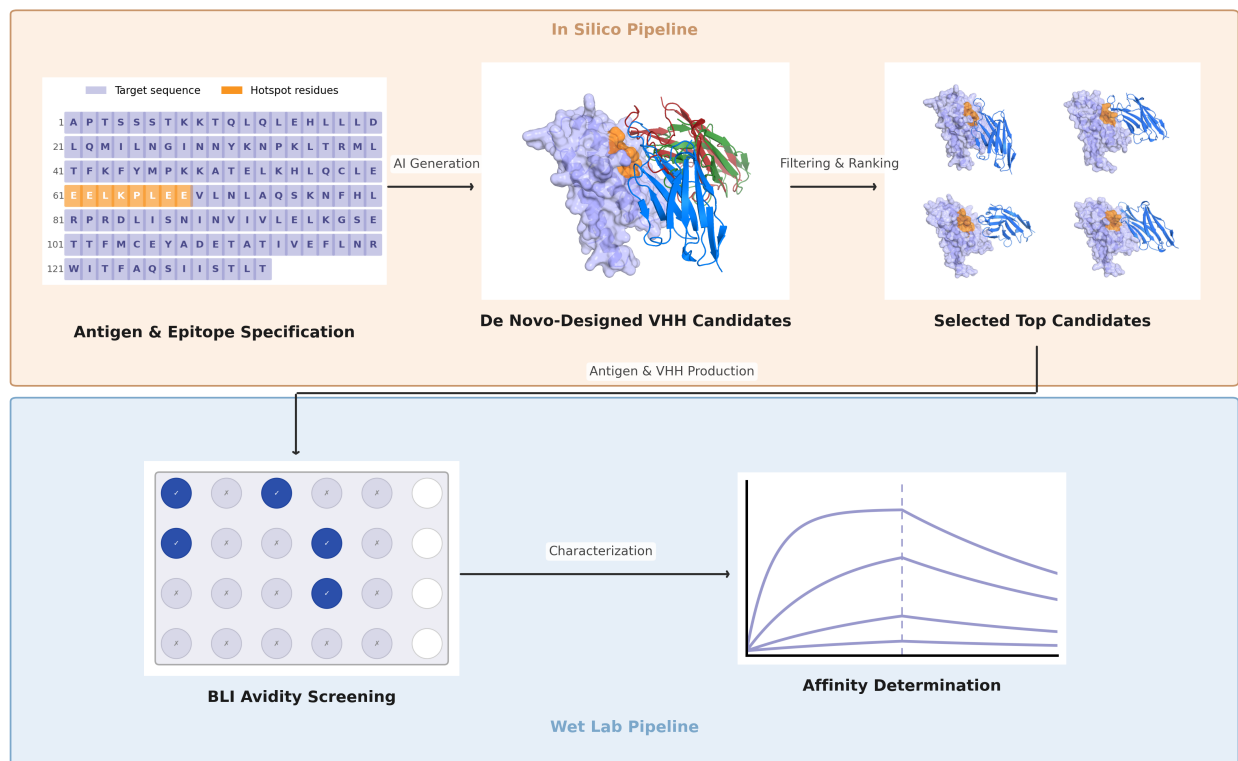


Figure 8: The MMDesign biologics discovery workflow. Candidate VHH sequences and structures are generated via hallucination-based optimization (**Design**), passed through multi-tier computational filtering (**Filter**), validated by BLI binding assays (**Experiment**), and characterized by kinetic analysis (**Analyze**).

objectives including epitope contact, interface quality, structural plausibility, and sequence naturalness, guiding the sequence toward regions of the confidence landscape that correspond to high-quality predicted complexes. Framework regions remain fixed throughout the optimization. Each campaign produces tens of thousands of candidate VHH sequences.

D.4 Computational filtering

The filtering cascade evaluates three successive criteria, each targeting a distinct failure mode:

Structural reliability. Reliability is assessed along four complementary axes: (a) confidence metrics from a single structure predictor (e.g. pLDDT, ipTM), (b) reproducibility across multiple random seeds of the same model, (c) concordance between two or more independent prediction architectures, and (d) epitope contact verification: in the predicted complex structure, CDR residues are checked to confirm direct contact with the specified hotspot positions. Designs that yield divergent conformations under seed or model perturbation are deprioritized.

Expressibility. A protein language model evaluates sequence naturalness, and a surface-property filter removes designs with elevated aggregation propensity.

Binding specificity. An in-house developed physics-based scoring function estimates binding free energy and evaluates whether the predicted interface exhibits features consistent with specific molecular recognition.

After all three tiers, the initial pool of tens of thousands is narrowed to **tens of designs** for wet-lab testing. When the number of candidates passing all filters exceeds the experimental budget, the same scoring function is used to rank and prioritize the final selection.

D.5 Experimental Characterization

Protein expression and purification. The CH2 and CH3 sequence of human IgG1 was fused to the C-terminus of each designed VHH. VHH-Fc constructs were expressed in a CHO system at 10 mL scale for 5 days; supernatant was collected and purified using Protein A magnetic beads. Purity was confirmed by SDS-PAGE under reducing conditions and SEC-HPLC.

Binding characterization. Multi-cycle kinetic assays were conducted by bio-layer interferometry (BLI) using a Gator Pivot instrument.

Primary binding hit screen. Target proteins and HSA, each carrying a His tag, were immobilized on separate NTA biosensors, and VHHs were applied as analyte at a single high concentration (2,000 nM). Candidates exhibiting binding to the target but not to HSA were classified as specific binders. The assay sequence was: 60 s baseline 1, 120 s loading (target shift 0.5 nm), 120 s baseline 2, 180 s association phase, 300 s dissociation phase, and 30 s regeneration and neutralization.

K_D determination. Hits from the primary screen were advanced to quantitative affinity measurement. VHHs were immobilized on HFC II biosensors via human IgG1 Fc tag, and target proteins were applied as analyte in four- to seven-point series of two-fold dilutions, enabling accurate kinetic fitting. The assay sequence was: 60 s baseline 1, 120 s loading (target shift 1.5 nm), 120 s baseline 2, 180 s association phase, 300 s dissociation phase, and 30 s regeneration and neutralization.

General conditions. All steps were performed at 25 °C with 10 mM HEPES, 150 mM NaCl, 3 mM EDTA, and 0.05% surfactant P20 (pH 7.4) as running buffer. Negative controls (buffer-only samples) were included for baseline subtraction. Data were collected at 10 Hz and fit globally to a 1:1 binding model across all concentrations to obtain K_D .

Poly-reactivity assessment. To confirm that binding was mediated by specific epitope interactions rather than non-specific electrostatic interactions, a poly-reactivity ELISA was performed using insulin and dsDNA as surrogate antigens, coated at 3 µg/ml at 4 °C overnight. After blocking with 2% BSA, VHHs were tested at two concentrations (5 µg/ml and 0.5 µg/ml) and incubated at 37 °C for 1 h. Detection was performed with HRP-conjugated anti-human IgG Fc antibody (1:5000), TMB substrate, and 2 M H₂SO₄ stop solution; absorbance was read at 450 nm.